

Industrial Internship Report on "Traffic Forecasting Using Machine Learning"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Traffic Forecasting Using Machine Learning". The objective of the project was to analyze traffic-related data and build predictive models capable of forecasting traffic volume using machine learning techniques. The project involved data preprocessing, feature engineering, exploratory data analysis, model training, and performance evaluation. Through this work, practical knowledge of Python, Pandas, NumPy, Scikit-Learn, data visualization, and machine learning workflows was gained.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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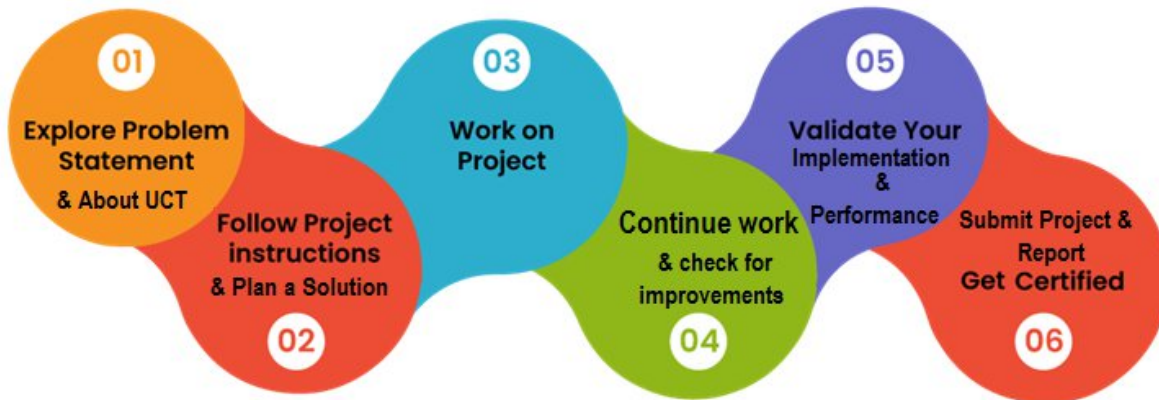
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1 Preface

This report presents the work carried out during the six-week Industrial Internship conducted by upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The internship provided practical exposure to industry-oriented problem solving and machine learning applications.

The assigned project, "**Traffic Forecasting Using Machine Learning**," focused on analyzing historical traffic data and developing predictive models to forecast future traffic volume. The internship enhanced my technical skills in Python, data preprocessing, feature engineering, model development, and performance evaluation.

The overall internship workflow followed the stages shown below, beginning with problem understanding and ending with project submission and certification.



The experience helped strengthen my understanding of real-world machine learning workflows and improved my problem-solving, analytical, and project documentation skills. I sincerely thank the mentors, upskill Campus, UCT, and my peers for their valuable guidance and support throughout the internship.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



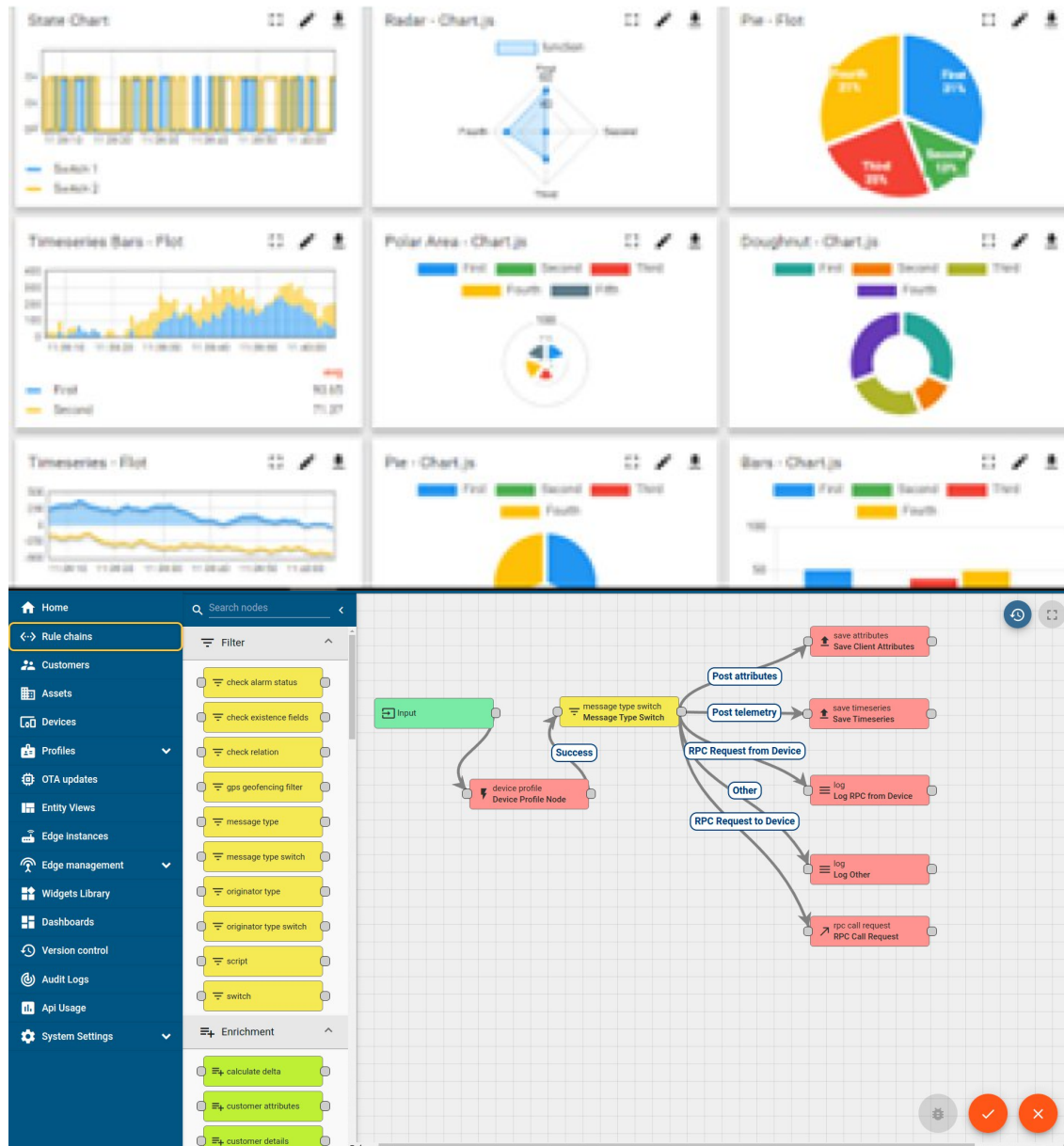
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



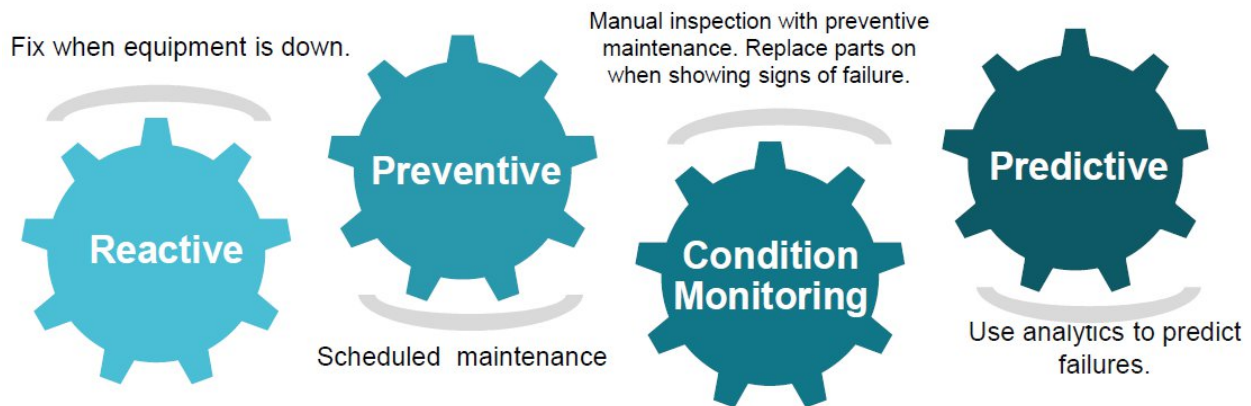


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

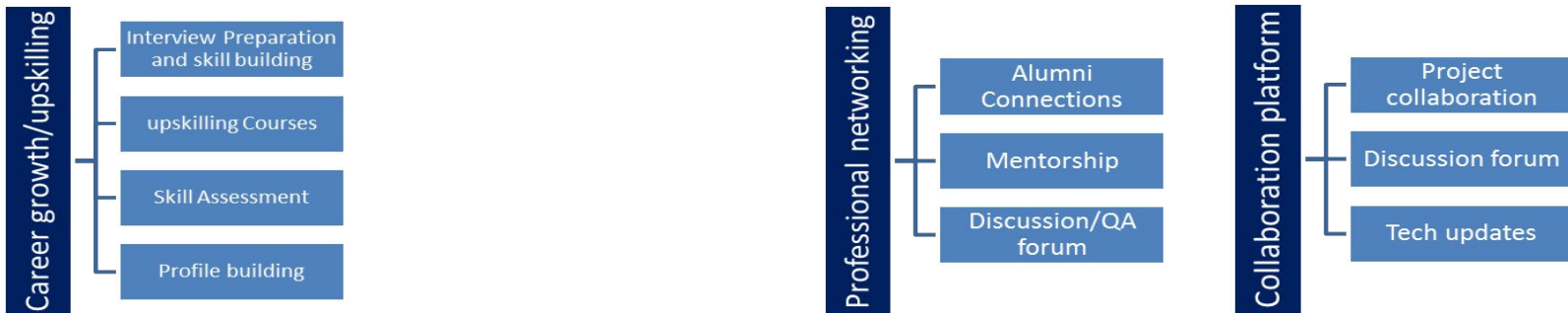
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.3 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.4 Reference

- [1] Scikit-learn Documentation: <https://scikit-learn.org>
- [2] Pandas Documentation: <https://pandas.pydata.org>
- [3] Traffic Forecasting Dataset, Kaggle/UCI Repository.

2.5 Glossary

Terms	Acronym
Machine Learning	ML
Random Forest Regressor	RF
Linear Regression	LR
Root Mean Square Error	RMSE
Mean Absolute Error	MAE
Dataset	DS
Feature Engineering	FE
Traffic Forecasting	TF
Training Dataset	Train Set
Testing Dataset	Test Set

3 Problem Statement

In the assigned problem statement

With the rapid growth of urbanization and increasing number of vehicles, traffic congestion has become a major challenge in modern cities. Traffic flow varies across different junctions and changes significantly based on time, day, and seasonal factors. Inefficient traffic management can lead to increased travel time, fuel consumption, and environmental pollution.

The objective of this project is to develop a machine learning-based traffic forecasting system that can predict future traffic volumes using historical traffic data. The dataset contains vehicle counts recorded at different city junctions over a period of time. By analyzing temporal patterns and traffic trends, the system aims to provide accurate traffic predictions that can assist traffic authorities in better planning and management.

The proposed solution uses machine learning techniques such as Linear Regression and Random Forest Regressor to forecast traffic patterns. The predictions generated by the system can support smart city initiatives, reduce congestion, improve transportation efficiency, and enable data-driven decision-making for urban traffic management.

4 Existing and Proposed solution

Existing Solution

Traditional traffic management systems primarily rely on manual monitoring, fixed traffic signal timings, and historical observations. While these methods help in managing traffic to some extent, they have several limitations:

- Inability to accurately predict future traffic congestion.
- Limited utilization of historical traffic data.
- Inefficient response to sudden traffic fluctuations.
- Increased travel delays during peak hours.
- Lack of data-driven decision-making capabilities.

Proposed Solution

The proposed Traffic Forecasting System uses Machine Learning techniques to predict future traffic volumes based on historical traffic data collected from city junctions. The system performs data preprocessing, feature engineering, and model training to identify traffic patterns and generate accurate predictions.

The solution utilizes Linear Regression and Random Forest Regressor models for traffic forecasting. Time-based features such as hour, day, month, and day of the week are extracted from the dataset to improve prediction accuracy.

Value Addition

- Improved traffic forecasting accuracy.
- Better traffic management and congestion reduction.
- Support for Smart City initiatives.
- Data-driven planning for transportation authorities.
- Enhanced road infrastructure utilization.
- Reduced travel time and improved commuter experience.

4.1 Code submission (Github link) :

<https://github.com/devikapolarapu/upskillcampus>

4.2 Report submission (Github link) :

https://github.com/devikapolarapu/upskillcampus/blob/main/Traffic_Forecasting_Devika_Polavarapu_USC_UCT.pdf

5 Proposed Design/ Model

The proposed Traffic Forecasting System uses Machine Learning techniques to predict future traffic volume based on historical traffic data collected from city junctions. The system begins by loading and preprocessing the dataset, followed by feature engineering to extract useful time-based attributes such as hour, day, month, and day of the week, while categorical junction data is transformed using one-hot encoding. The processed dataset is then divided into training and testing sets, and two machine learning models, Linear Regression and Random Forest Regressor, are trained using the training data. Their performance is evaluated on the test dataset using RMSE (Root Mean Square Error) and MAE (Mean Absolute Error) metrics, and the model with better predictive performance is selected to generate accurate traffic forecasts, enabling data-driven decision-making for improved traffic management and transportation planning.

5.1 High Level Diagram

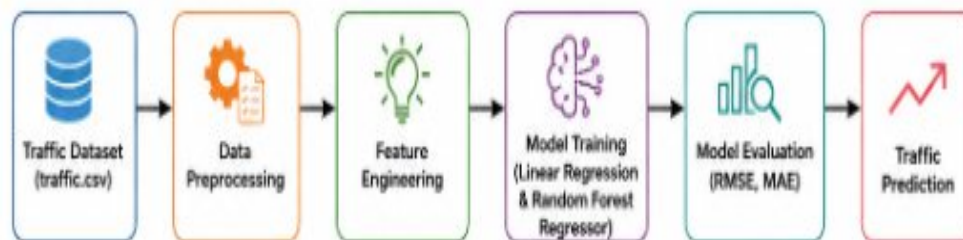


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

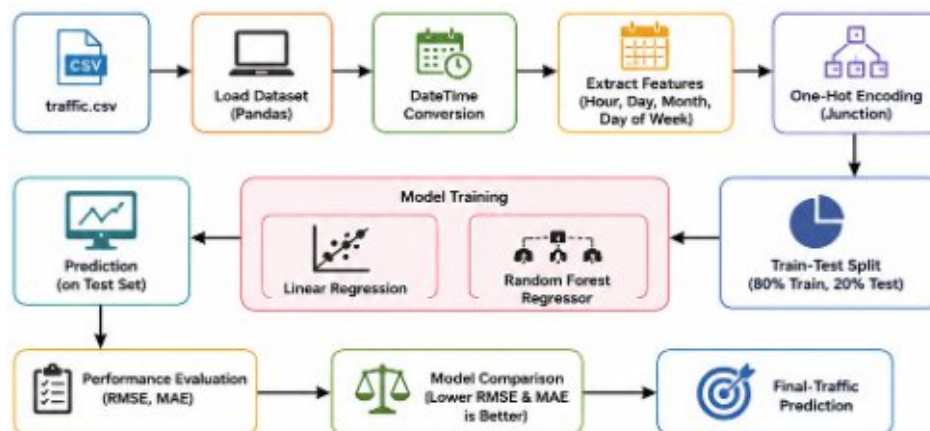


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

6 Performance Test

The proposed Traffic Forecasting system was evaluated using real traffic data. Linear Regression and Random Forest models were trained and tested using an 80:20 train-test split. RMSE and MAE metrics were used to measure prediction accuracy and compare model performance.

6.1 Test Plan/ Test Cases

Test Case ID	Test Description	Expected Result	Status
TC-01	Load traffic dataset	Dataset loaded successfully	Pass
TC-02	Convert DateTime column	Date and time features extracted	Pass
TC-03	Apply one-hot encoding on Junction	Categorical data converted to numerical format	Pass
TC-04	Train Linear Regression model	Model trained successfully	Pass
TC-05	Train Random Forest model	Model trained successfully	Pass
TC-06	Generate predictions	Traffic volume predictions generated	Pass
TC-07	Evaluate model performance	RMSE and MAE calculated	Pass

6.2 Test Procedure

1. Loaded the traffic dataset using Pandas.
2. Performed data preprocessing and DateTime conversion.
3. Extracted time-based features such as hour, day, month, and day of week.
4. Applied one-hot encoding to the Junction feature.

5. Split the dataset into training (80%) and testing (20%) sets.
6. Trained Linear Regression and Random Forest Regressor models.
7. Generated predictions on the test dataset.
8. Evaluated model performance using RMSE and MAE metrics.
9. Compared both models and selected the better-performing model.

6.3 Performance Outcome

The Traffic Forecasting system successfully predicted vehicle traffic using historical traffic data. Both Linear Regression and Random Forest Regressor models were evaluated using RMSE and MAE metrics.

The Random Forest Regressor achieved better prediction accuracy compared to Linear Regression due to its ability to capture complex patterns in traffic data.

The implemented solution demonstrates the effectiveness of machine learning techniques for traffic forecasting and can assist transportation authorities in traffic management, congestion reduction, and future planning.

7 My learnings

During this internship, I gained practical experience in applying Machine Learning techniques to real-world datasets. I learned data preprocessing, feature engineering, model training, and performance evaluation using Python and Scikit-learn. I also understood the importance of selecting appropriate evaluation metrics such as RMSE and MAE for regression problems.

Additionally, I improved my problem-solving skills, GitHub project management, documentation practices, and gained exposure to industry-oriented project development workflows.

8 Future work scope

The Traffic Forecasting system can be further enhanced by using advanced deep learning models such as LSTM and GRU for improved time-series forecasting accuracy. Additional features such as weather conditions, special events, and holiday information can also be incorporated to improve predictions.

The system may be integrated with real-time traffic monitoring systems and smart city infrastructure to support dynamic traffic management and congestion control.